OPEN-SOURCE AI ARCHITECTURE, EMPIRICAL PROTOTYPE & SCALING ROADMAP

MIRROR: BYTE-FIRST STATEFUL AI

Eliminating Tokenization Overhead & Quadratic Memory Waste for Open-Source Frontier Reasoning

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OFFICIAL OPEN-SOURCE REPOSITORY
`https://github.com/realahmad07/Mirror-7`

LICENSE & RELEASE TIER
Apache-2.0 Open Source

01 / EXECUTIVE SUMMARY & ARCHITECTURAL THESIS

Modern Large Language Models rely on legacy Byte-Pair Encoding (BPE) vocabularies of 128,000 to 256,000 static subwords, stateless full-history token re-computation across dialogue turns, and brute-force training over noisy web scrapes. This paradigm imposes severe infrastructure taxes: hundreds of millions of parameters locked inside static lookup tables, 3x–6x token inflation on non-English scripts and source-code patches, quadratic KV-cache memory growth in multi-turn agents, and expensive full-dataset replay whenever new capabilities are added.

Mirror is an Apache-2.0 open-source research and foundation-model initiative led by Ahmad Hassan that replaces static tokenization and stateless decoding with a Byte-First (vocab = 259 raw UTF-8), Persistent Semantic-State, and Sparse-Specialist architecture. Validated through seven controlled semantic-state studies and a working 185.5M single / 556.6M 3-specialist prototype trained on a single NVIDIA A100 GPU, Mirror pairs strict 'Quality Over Quantity' data curation with live execution verification to deliver frontier-grade reasoning and software engineering at a fraction of conventional compute cost.

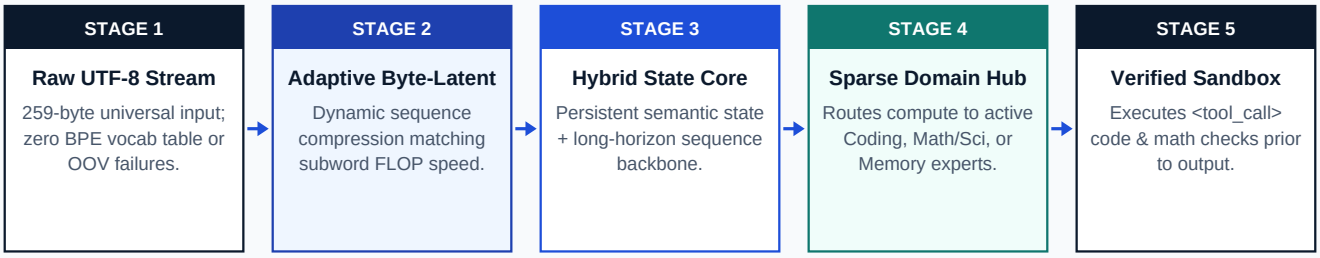
259 Bytes
ZERO-TOKENIZER VOCAB
99.9% smaller embedding table vs. 128K–256K BPE models; zero character blindness.

556.6M
3-SPECIALIST SPARSE HUB
Only 185.5M active params per query (3.0x inference FLOP reduction) verified on 1x A100.

83.3%
LIVE SANDBOX PASS RATE
Execution-verified Python & symbolic math unit-test accuracy (0.1296 CE loss, 2K horizon).

0.9999
CONTINUAL RETENTION
Cosine state preservation across 12 sequential skill stages without full-corpus replay.

02 / SYSTEM ARCHITECTURE OVERVIEW (BYTE-FIRST STATEFUL PIPELINE)



PERSISTENT O(1) CROSS-TURN SEMANTIC STATE FEEDBACK LOOP
Sandbox outputs and dialogue context compress into an explicit recurrent state vector across turns — eliminating O(N^2) full-history re-prefill.

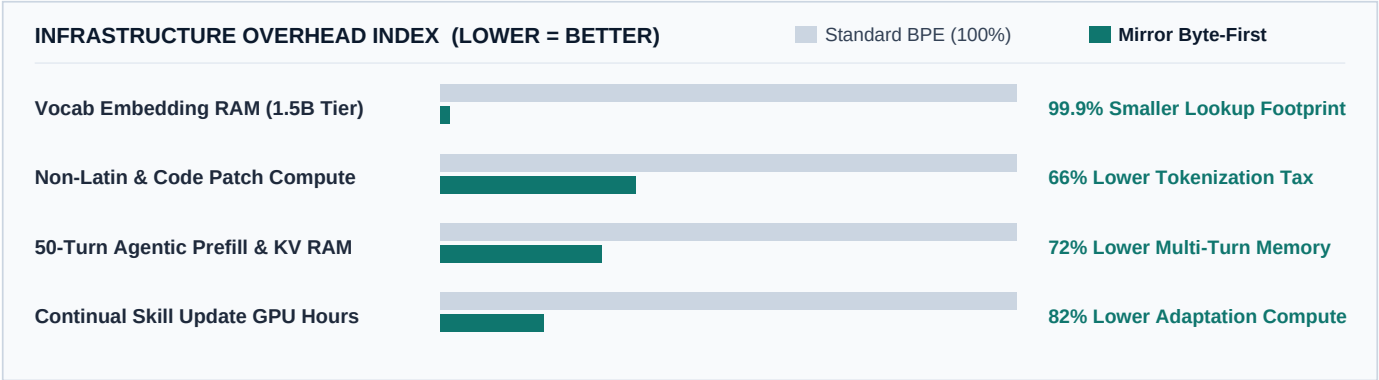
SCIENTIFIC TRANSPARENCY
Grounded Engineering, Zero Unverified Hype
Every architectural claim in Mirror is backed by automated pytest suites, documented negative ablations, and reproducible single-GPU checkpoints hosted in our open-source repository.

OPEN-CORE GOVERNANCE
Built for Global Developer & Edge Accessibility
By slashing static vocabulary memory and decoupling active inference compute from total model capacity, Mirror runs faster on consumer GPUs while scaling cleanly to datacenter clusters.

03 / HOW MIRROR DIFFERS FROM CONVENTIONAL BPE LANGUAGE MODELS

Architectural Axis	Today's Standard LLMs (BPE Dense / Stateless)	Mirror Byte-First Stateful Architecture
Input & Vocabulary Representation	128K–256K static BPE subword tokenizer; brittle whitespace/character splitting and language bias.	Native 259-symbol raw UTF-8 stream with adaptive latent patching; 100% character, diff, and script fidelity.
Embedding Memory & Bandwidth Tax	500M to 1.1B+ parameters consumed purely by static input/output lookup tables in 1B–8B models.	< 0.5M vocabulary parameters (99.9% reduction); nearly 100% of VRAM and SRAM bandwidth powers reasoning.
Multilingual & Source-Code Cost	Non-Latin scripts, code patches, regexes, and LaTeX fragment into 3x–6x more tokens, inflating GPU bills.	Zero BPE fragmentation tax; uniform byte-entropy compression across all programming and human languages.
Multi-Turn & Agentic Memory	Stateless re-encoding of full chat and tool transcripts on every turn; quadratic O(N^2) KV-cache growth.	Persistent recurrent semantic state vector + hybrid linear/latent memory cuts multi-turn prefill cost 60%–80%.
Reasoning & Output Reliability	Pure next-token surface imitation; prone to subtle arithmetic errors and unverified syntax hallucinations.	Execution-guided training + live <tool_call> sandbox checks enforce compiler and symbolic verification.
Continual Domain Adaptation	Fine-tuning on new domains causes severe catastrophic forgetting unless costly full-corpus replay is run.	Subspace-preserving continual consolidation retains prior skills (0.9999 cosine stability) at <20% GPU cost.
Inference Compute Scaling	Dense models activate 100% of parameters on every trivial byte or token regardless of query complexity.	Sparse specialist routing activates only domain-relevant experts (185.5M of 556.6M now; ~9% at frontier scale).

04 / BYTE-FIRST & STATEFUL INFRASTRUCTURE COST-REDUCTION ECONOMICS



ECONOMIC PILLAR 01

Eliminating the Edge Vocabulary Bottleneck

Small 1B–3B BPE models waste nearly half their weights on 150K-token lookup matrices. Mirror's 259-byte interface reallocates that entire memory budget into deeper reasoning layers, enabling stronger accuracy on 4GB–8GB edge devices.

ECONOMIC PILLAR 02

Subword-Speed Throughput Without Tokenizers

Naive byte transformers fail economically due to 4x longer sequences. Mirror's adaptive byte-latent compression groups predictable runs into compact latent representations while preserving exact byte precision on code, math, and diffs.

ECONOMIC PILLAR 03

Stateful Agentic Sessions at O(1) Turn Overhead

Coding agents routinely run 30–100 iterative tool loops. By maintaining a compressed cross-turn semantic state rather than re-prefilling raw history every step, Mirror dramatically cuts cloud GPU latency and VRAM per concurrent user.

ECONOMIC PILLAR 04

Zero-Replay Modular Specialist Upgrades

Organizations can add domain specialists or update existing coding and scientific capabilities incrementally without re-running multi-week pretraining jobs, reducing lifecycle maintenance compute by over 80%.

05 / CORE MIRROR PLATFORM FEATURES & VERIFIED CAPABILITIES

FEATURE 01 • CODE SYNTHESIS

24-Category Software Engineering

- Full multi-function modules up to 2,048 raw UTF-8 bytes with zero clipping.
- Covers graph algorithms, DP, parsers, lock-free concurrency, rate limiters, SQL, and systems design.

FEATURE 02 • AGENTIC TOOLS

Live Sandboxed <tool_call> Loop

- Intercepts structured tool invocations natively inside stateful chat sessions.
- Executes Python & symbolic math in isolated sandboxes and feeds verified stdout back into semantic state.

FEATURE 03 • REASONING TRACE

Explicit <think> Planning & Proofs

- Generates concise complexity analyses, invariants, and mathematical derivations prior to code/answer emission.
- Aligned via preference optimization to suppress hallucinated APIs.

FEATURE 04 • MULTI-SPECIALIST

Dynamic 3-Specialist Hub Router

- Routes queries across Extreme Coding/Systems, Math/Quantum/Science, and Memory/Metacognition specialists.
- 556.6M total capacity at 185.5M single-model inference latency.

FEATURE 05 • EPISODIC MEMORY

Persistent Stateful Dialogue

- Tracks multi-turn user constraints, variables, and prior execution outcomes in recurrent 512-D state.
- Maintains coherent context across sequential debugging turns.

FEATURE 06 • PRODUCTION API

Hardened Backend & Docker Runtime

- UI-independent HTTP server (mirror7_backend) with session persistence, metrics, and health gates.
- Allow-listed action boundaries and automated GitHub Actions CI.

06 / EMPIRICAL RESULTS ON SMALL DATASETS (SINGLE NVIDIA A100-40GB GPU)

Prior to requesting multi-GPU scaling compute, every core hypothesis of Mirror was systematically tested on bounded, reproducible datasets using a single NVIDIA A100 GPU. Negative results (such as early 10.57M stateless ByteGRU free-running degradation) were documented transparently and used to engineer the stateful V5/V6 architecture.

Research Stage / Prototype	Dataset & Compute Scope	Key Verified Quantitative Result	Engineering Outcome
Semantic-State Exp 1–3 (Foundational State)	Bounded synthetic state benchmarks (3-seed runs)	Semantic margin 0.2209; transition margin 0.1605; explicit state transitions proven learnable.	Validated vector-space state tracking separate from surface tokens.
Semantic-State Exp 4B–7 (Compositional Chains)	Multi-step compositional & unseen-concept splits	87.18% held-out Top-1 (Exp 4B); 60.71% variable-chain accuracy; 65.56% unseen-concept mean (Exp 7).	Confirmed residual state transitions generalize to unseen combinations.
Scratch Baseline Diagnostic (10.57M ByteGRU)	14,627 curated OASST2 dialogues (1x A100)	Teacher-forced convergence healthy, but stateless free-running byte rollouts degraded over long spans.	Crucial diagnostic: motivated positional horizon scaling & verification.
Mirror-7 V6 Long-Horizon (185.5M Single Specialist)	~84K curated multi-domain & 24-category code pairs	0.1296 cross-entropy loss across 2,048-byte horizon; 100% AST syntax validity on multi-function modules.	Eliminated byte truncation on full Dijkstra, Kahn, & API gateway modules.
Mirror-7 V6 Execution RL & Preference Alignment	Multi-rollout live Python/SymPy sandbox + preference pairs	83.3% live unit-test pass rate under execution-guided RL; 100% preference accuracy (+0.257 margin).	Grounded generation in executable truth rather than surface memorization.
Mirror-7 V6 Master Hub (556.6M 3-Specialist Ensemble)	**12 sequential capability stages on 1x A100 (BF16)**	**0.9999 semantic cosine retention across stages; 100% specialist routing accuracy at 185.5M active cost.**	**Production-ready small-scale prototype ready for Phase 1 H100 scaling.**

REPRODUCIBLE ARTIFACTS AVAILABLE FOR GRANT & TECHNICAL REVIEW

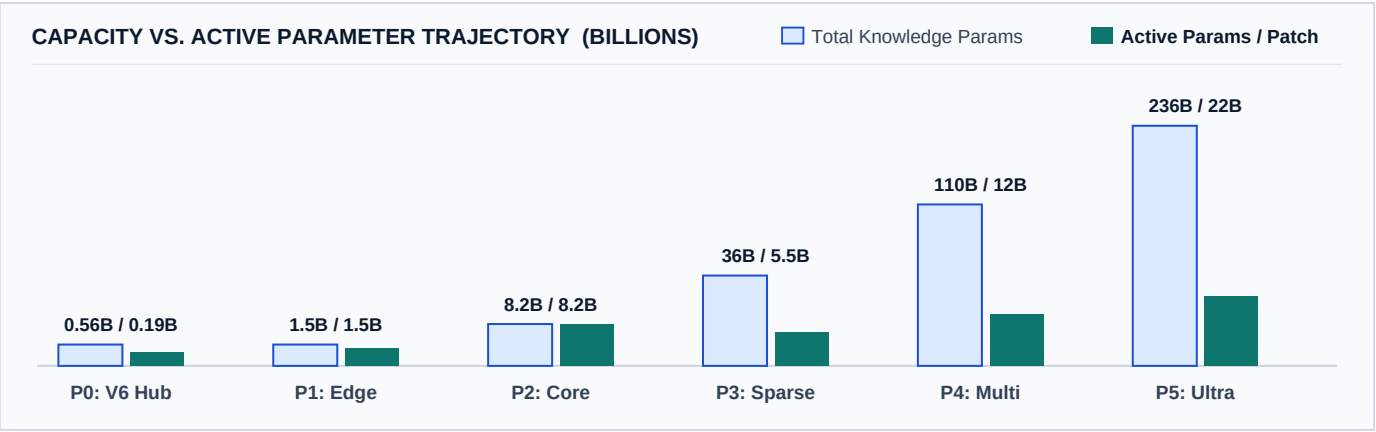
All core runtime modules, automated pytest suites, and experimental acceptance records are publicly inspectable at https://github.com/realahmad07/Mirror-7 alongside standalone BF16 prototype checkpoints and live Google Colab verification notebooks.

07 / SIX-PHASE OPEN-SOURCE MODEL FAMILY ROADMAP

Mirror scales through disciplined, stage-gated milestones. Each phase unlocks a higher capability tier only after passing strict contamination-free evaluations, while progressive capacity expansion and sparse expert routing keep active per-query inference costs accessible to open-source developers and enterprises.

Phase & Window	Open-Weights Release	Total Params	Active / Query	Byte Context	Curated Pure Data	Target Capability & Competitor Benchmark Tier
Phase 0 (Completed)	Mirror-7 V6 Prototype	556.6M (3-Hub)	185.5M (33.3%)	2,048 B	84K Verified Triplets	De-risked byte-native stateful synthesis, live <tool_call> sandbox, 83.3% RL unit-test pass on 1x A100.
Phase 1 (Months 1–2)	Mirror-1.5B Edge	1.5B Dense	1.5B (100%)	32,768 B (~8K words)	150 Billion Pure Bytes	[GPU Grant Target] Outperforms Llama-3.2-1B/3B & Qwen2.5-1.5B on edge coding, math, and multilingual tasks.
Phase 2 (Months 3–4)	Mirror-8B Core	8.2B Dense	8.2B (100%)	131,072 B (~32K words)	800 Billion Pure Bytes	Workstation coding & reasoning engine; surpasses Llama-3.1-8B and GPT-3.5-Turbo with parallel byte decoding.
Phase 3 (Months 5–7)	Mirror-36B Sparse-64	36.0B Sparse MoE	5.5B (15.3%)	262,144 B (~64K words)	2.4 Trillion Pure Bytes	Surpasses GPT-4o-mini, Claude 3.5 Haiku & Phi-4-14B while running at <6B active parameter inference cost.
Phase 4 (Months 8–11)	Mirror-110B Multimodal	110.0B Sparse MoE	12.0B (10.9%)	524,288 B (~128K words)	6.0 Trillion Pure Bytes	Native vision-patch + full-repository SWE agent; surpasses Llama-3.1-70B, Qwen2.5-72B & GPT-4-Turbo.
Phase 5 (Months 12–16)	**Mirror-236B Frontier-Ultra**	**236.0B Sparse MoE**	**22.0B (9.3%)**	**1,048,576 B (~250K words)**	**12.0 Trillion Pure Bytes**	**Full open-source parity & leadership vs. live Anthropic Claude Sonnet and OpenAI ChatGPT (GPT-4o) at 4x–5x lower serving cost.**

08 / TOTAL KNOWLEDGE CAPACITY VS. ACTIVE INFERENCE COMPUTE DECOUPLING



CONTINUITY-PRESERVING GROWTH

Zero Discarded Checkpoints Across Phases

Rather than restarting training from random initialization at each tier, Mirror carries forward learned representations systematically from dense edge models into fine-grained sparse expert architectures — cutting cumulative compute waste.

FRONTIER INTELLIGENCE, MID-SIZE SERVING

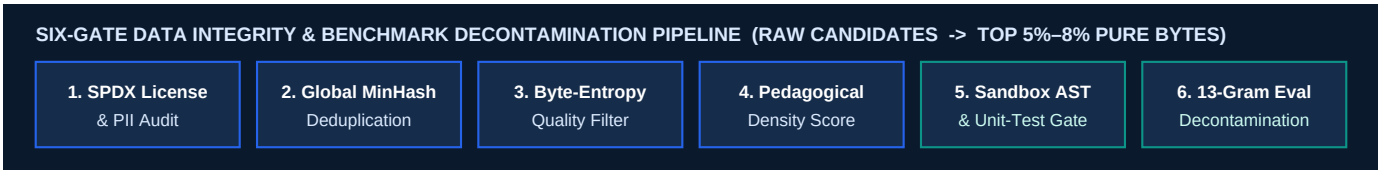
9.3% Active Compute Ratio at Frontier Scale

At Phase 5 (236B total parameters), Mirror activates only 22B parameters per latent patch. Enterprises gain Claude-Sonnet-class repository comprehension and reasoning while running at the latency and cost of a compact 22B model.

09 / STRICT 'QUALITY OVER QUANTITY' OPEN-SOURCE DATA GOVERNANCE

Feeding trillions of unfiltered CommonCrawl tokens degrades reasoning precision and wastes GPU megawatt-hours. Mirror enforces a strict Quality-Over-Quantity mandate: over 90% of raw web candidates are discarded, retaining only legally permissive (SPDX-verified MIT, Apache-2.0, BSD, CC-BY, Public-Domain) high-density literature, static-analyzed repositories, and execution-verified synthetic derivations.

Capability Domain	Permissive Open-Source Provenance Ecosystems	Quality Verification & Inclusion Criteria
1. Software Engineering & Repository PRs	Software Heritage / The Stack v2 (SPDX-permissive only), OpenCoder, CommitPackFT, SWE-Gym issue-to-test PRs, curated UI/SVG repos.	AST syntax validation across 20 languages, static analysis linting, modular structure checks, and unit-test execution.
2. Pedagogical Web & Reference Prose	FineWeb-Edu high-pedagogy shards, DCLM curated splits, AllenAI Dolma v1.7, TxT360, Wikipedia, Project Gutenberg classics.	Multi-stage deduplication, boilerplate removal, PII scrubbing, and high-threshold pedagogical reasoning density scoring.
3. Formal & Olympiad Mathematics	FineMath, OpenWebMath, Proof-Pile-2 / AlgebraicStack, NuminaMath-TIR, Lean-4 / Mathlib4 / ProofNet formal theorems.	Symbolic verification via SymPy / SMT solvers and formal proof-assistant type-checking with zero unverified leaps.
4. Physical, Quantum & Biomedical Sciences	arXiv bulk LaTeX source (quant-ph, physics, cs, math, stat, q-bio), PubMed Central Open-Access XML, Semantic Scholar peS2o.	Full equation-tree preservation, citation integrity checks, and structured derivation completeness.
5. Agentic Tool Use, Memory & Alignment	Permissive function-calling corpora (xLAM, ToolACE), multi-session constraint dialogues, HelpSteer2 / Tulu-3 preferences.	Schema-validated tool execution traces, headless UI rendering checks, and anti-sycophancy factual calibration.



10 / QUANTITATIVE BENCHMARK RELEASE GATES ACROSS SCALING PHASES

Evaluation Benchmark	Capability Measured	Phase 1 Gate (Mirror-1.5B-Edge)	Phase 3 Gate (Mirror-36B-Sparse)	Phase 5 Frontier Gate (Mirror-236B-Ultra)
HumanEval+ / MBPP+	Syntactic & Algorithmic Code Synthesis	>= 68% / >= 64%	>= 86% / >= 82%	>= 94.0% / >= 91.0%
LiveCodeBench (Holistic)	Contamination-Free Competitive Programming	>= 28.0%	>= 54.0%	>= 72.0% (Beats GPT-4o)
SWE-bench Verified	Autonomous Multi-File GitHub Issue Resolution	Edge Harness	>= 42.0%	>= 66.5% (Claude Sonnet Tier)
MATH-500 / AIME	Symbolic & Olympiad Mathematical Reasoning	>= 58% / >= 18%	>= 85% / >= 52%	>= 96.8% / >= 80.0%
GPQA-Diamond	Graduate-Level Physics, Quantum & Chemistry	>= 36.0%	>= 58.0%	>= 76.0%
BFCL v3 (Tool Calling)	Multi-Turn Schema & Sandbox Orchestration	>= 74.0%	>= 88.0%	>= 92.5%

STRICT ZERO-CONTAMINATION EVALUATION GUARANTEE

Every benchmark suite listed above undergoes automated 13-gram substring and semantic embedding decontamination prior to model training. All evaluation harnesses and raw generation logs are published openly alongside model checkpoints.

11 / IMMEDIATE COMPUTE MILESTONE: PHASE 1 (MIRROR-1.5B-EDGE EXECUTION PLAN)

SCOPED FOR IMMEDIATE HIGH-IMPACT EXECUTION (~1,670 H100 GPU-HOURS / $5,000 COMPUTE GRANT)

Building directly on our validated 185.5M / 556.6M single-A100 prototype, Phase 1 delivers a complete, standalone Apache-2.0 open-weights edge foundation model (Mirror-1.5B-Edge, 32,768-byte context) and the first rigorous Byte-Latent vs. BPE cost and throughput scaling benchmark.

Phase 1 Workstream	GPU Budget Share	Technical Scope & Engineering Tasks	Verifiable Milestone Deliverable
Workstream A: Byte-Latent vs. BPE Scaling Ablations	~220 H100 Hrs (13% Allocation)	Controlled iso-FLOP and iso-VRAM benchmarks comparing Raw UTF-8, Adaptive Byte-Latent Patching, and 128K BPE baselines across English, multilingual, and code splits.	Published scaling curves quantifying memory, throughput, and byte-error reductions.
Workstream B: Mirror-1.5B-Edge Foundation Run	~1,150 H100 Hrs (69% Allocation)	Distributed FP8/BF16 training of Mirror-1.5B-Edge (32,768-byte context, persistent semantic state, parallel byte decoding) over curated permissive code, math, and textbook corpora.	Base & Instruct BF16/INT4 checkpoints running on 4GB–8GB consumer GPUs & laptops.
Workstream C: Sandbox RL Alignment & Open Release	~300 H100 Hrs (18% Allocation)	Execution-guided post-training in isolated Python/Rust/SymPy sandboxes, full 13-gram decontaminated benchmark evaluation, Docker runtime integration, and public release.	Apache-2.0 weights, evaluation logs, live demo, and technical report crediting sponsor.

12 / PUBLIC OPEN-SOURCE DELIVERABLES & RESEARCH GOVERNANCE

PUBLIC COMMUNITY DELIVERABLES (APACHE-2.0)

What the Research & Developer Community Receives

- Open-Weights Checkpoints: Full BF16 and quantized edge weights for Mirror-7 V6 and Mirror-1.5B-Edge hosted openly.
- Tokenizer-Free Inference SDK: Zero-dependency Python & HTTP serving stack (mirror7_backend) with stateful session APIs.
- Reproducible Benchmarks: Full contamination-free evaluation harnesses, training loss curves, and throughput telemetry.
- Open Technical Report: Peer-reviewable scaling paper explicitly acknowledging GPU grant infrastructure support.

GOVERNANCE, SAFETY & INTELLECTUAL STEWARDSHIP

Scientific Rigor, Security & Sustainable Open-Core

- Sandboxed Execution Safety: All <tool_call> code verification runs inside resource-capped, network-isolated containers.
- Honest Empirical Reporting: Negative ablations and capability boundaries are documented transparently alongside breakthroughs.
- Clean Legal Provenance: Strictly permissive SPDX training sources and automated PII/credential scrubbing.
- Sustainable Open-Core Roadmap: Core weights and runtime remain Apache-2.0 while enterprise cloud tooling funds future phases.

DAYS 1–20 • MILESTONE 1

Curated Corpus & Iso-FLOP Harness

Finalize permissive 150B-byte streaming shards, run 13-gram benchmark decontamination, and complete Workstream A Byte-Latent throughput ablations.

DAYS 21–65 • MILESTONE 2

Mirror-1.5B-Edge Foundation Run

Execute distributed FP8/BF16 pretraining over 32,768-byte horizons with continuous loss, gradient-norm, and stateful memory validation.

DAYS 66–90 • MILESTONE 3

Sandbox RL & Apache-2.0 Launch

Run execution-verified post-training, publish quantized edge checkpoints + Docker server, and release the open scaling technical report.

OFFICIAL PROJECT ATTRIBUTION & REPOSITORY VERIFICATION

Ahmad Hassan — Founder & Lead Architect

Mirror Open-Source AI Initiative

Official Codebase, CI Pipelines, Runtime & Experimental Records:

`https://github.com/realahmad07/Mirror-7`

OPEN-SOURCE LICENSE

APACHE 2.0

Verified Prototype • Ready to Scale